

Guobin Zhao

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EDUCATION

Pusan National University

Ph.D. in Chemical Engineering, GPA: 3.96/4.50

Busan, KR

2020–2025

- Thesis: “Accelerating Nanoporous Materials Discovery for Adsorptive Separations: from Database to Machine Learning Models”

Jimei University

M.S. in Naval Architecture and Ocean Engineering

Xiamen, CN

2018–2020

- Thesis: “Study on Adsorption Storage of Methane on Typical MOFs”

Zhejiang Ocean University

B.S. in Marine Engineering (Ship Building and Repairing)

Zhoushan, CN

2014–2018

- Thesis: “Design of a Stacked-Type Hatch Cover”

EXPERIENCE

National University of Singapore

Postdoctoral Research Fellow at MACE Group

Singapore, SG

2025.12-2026.12

- Fine-tuning LLM for Validation of MOFs
- Validate if LLM understand crystal structures, and give the reason for unphysical structures.
- Graph Transformer - Based Method for Solvent Removal of MOFs
- Develop a machine learning model for free & coordinated solvent removal for MOFs.
- Refinement of MOF Structures by Generative AI
- Explore the application of generative AI in refining MOF structures.
- CO_2 Conversion by MOF
- Explore performance of CO_2 conversion by CALF-20 by DFT simulation.

Pusan National University

Postdoctoral Associate at Chung Research Group

Busan, KR

2025.09-2025.11

- CoRE MOF 2026 Database (Co-work with Prof. Kulik Group at Massachusetts Institute of Technology and Prof. Moosavi Group at University of Toronto.)
- Updated the CoRE MOF database and investigated the sensitivity of not computation-ready structures on machine learning models.
- Machine Learning Potential for MOFs (Co-work with Prof. Rosen Group at Princeton University.)
- High-throughput density functional theory calculations and training machine learning force fields.
- Zn-MOF for Direct Air Capture (Co-work with Prof. Koh Group at KAIST.)
- Found stable structure with high performance for carbon capture.

- Quantum Machine Learning Discovery of Nanoporous Materials for Direct Air Carbon Capture (Co-work with Prof. Moosavi Group at University of Toronto.)
- Analyzed the structures that simulation fail.

PUBLICATIONS

- [1] Y. Chen, J. Liu, **G. Zhao**, Z. Wang, C. Chen, J. Huang, X. Tian, and L. Huang, “Physically Verifiable Evidence and LLM-Based Reporting for Bearing Fault Diagnosis”, *under review*, 2026.
- [2] **G. Zhao** and X. Y. Li, “Chemically meaningful textualization enables explainable quality control of metal-organic frameworks for reliable functional screening”, *under review*, 2026.
- [3] **G. Zhao**, K. Liang, and X. Y. Li, “Hstability-aware structural cleaning of metal–organic frameworks for reliable molecular simulation and high-throughput screening”, *under review*, 2026.
- [4] J. Guo, X. Xu, L. E, **G. Zhao**, and Y. Li, “Beyond disappearance: Heterojunction aops and the hidden carbon fate in water systems”, *ACS ES&T Water*, 2026.
- [5] Y. Liu, Z. Yang, Z. Deng, C. Cao, Y. Liu, H. Peng, B. Poczos, J. Guo, Y. Chen, Z. Ming, **G. Zhao***, and P. S. Yu., “How can artificial intelligence accelerate the in silico screening of metal-organic frameworks?”, *under review*, 2026.
- [6] Y. Liu, Z. Yang, D. Guo, K. Peng, F. Yuan, C. Cao, C. Li, Z. Ma, Y. Liu, and **G. Zhao***, “Mof-sleuth: Tool-grounded reward alignment for explainable fine-grained mof cif auditing”, *under review*, 2026.
- [7] Y. Zhu, Y. Yang, **G. Zhao**, X. Y. Li, X. Wang, and P. Ou, “Literature-mining-guided design of lithium-functionalized polar penta-BCN nanosheets for reversible hydrogen storage”, *under review*, 2026.
- [8] Y. Chen, **G. Zhao**, L.-C. Lin, and Y. G. Chung, “A Master Isotherm Model Approach to Quantify Defects in UiO-66 from Nitrogen Adsorption Isotherms”, *Langmuir*, vol. 41, no. 44, pp. 29 851–29 858, 2025.
- [9] **G. Zhao**, L. M. Brabson, S. Chheda, J. Huang, H. Kim, K. Liu, K. Mochida, T. D. Pham, G. G. Terrones, S. Yoon, et al., “CoRE MOF DB: A Curated Experimental Metal–Organic Framework Database with Machine–Learned Properties for Integrated Material–Process Screening”, *Matter*, vol. 8, no. 6, 2025.
- [10] **G. Zhao**, P. Zhao, and Y. G. Chung, “MOFClassifier: A Machine Learning Approach for Validating Computation-Ready Metal–Organic Frameworks”, *Journal of the American Chemical Society*, vol. 147, no. 37, pp. 33 343–33 349, 2025.
- [11] S. Yoon, J. Jung, **G. Zhao**, Y. Lee, K. B. Lee, and Y. G. Chung, “Techno-Economic Analysis of Cu (I)–Loaded Porous Carbons for CO Separation via Pressure/Vacuum Swing Adsorption Process”, *Separation and Purification Technology*, vol. 378, no. 3, p. 134 786, 2025.
- [12] Y. Chen, **G. Zhao**, S. Yoon, P. Habibi, C. S. Hong, S. Li, O. A. Moulton, P. Dey, T. J. Vlugt, and Y. G. Chung, “Computational Exploration of Adsorption–Based Hydrogen Storage in Mg-Alkoxide Functionalized Covalent–Organic Frameworks (COFs): Force–Field and Machine Learning Models”, *ACS applied materials & interfaces*, vol. 16, no. 45, pp. 61 995–62 009, 2024.
- [13] Y. C. Chung, C. W. Yang, and **G. Zhao**, “Global Warming Potential (GWP) Prediction Apparatus based on Computational Chemistry and Machine Learning”, 10-2023-0061292, Patent, 2024.
- [14] **G. Zhao** and Y. G. Chung, “PACMAN: a Robust Partial Atomic Charge Predictor for Nanoporous Materials Based on Crystal Graph Convolution Networks”, *Journal of Chemical Theory and Computation*, vol. 20, no. 12, pp. 5368–5380, 2024.
- [15] **G. Zhao**, H. Kim, C. Yang, and Y. G. Chung, “Leveraging Machine Learning to Predict the Atmospheric Lifetime and the Global Warming Potential of SF₆ Replacement Gases”, *The Journal of Physical Chemistry A*, vol. 128, no. 12, pp. 2399–2408, 2024.

- [16] **G. Zhao**, Y. Chen, and Y. G. Chung, “High-Throughput, Multiscale Computational Screening of Metal–Organic Frameworks for Xe/Kr Separation with Machine–Learned Parameters”, *Industrial & Engineering Chemistry Research*, vol. 62, no. 37, pp. 15 176–15 189, 2023.
- [17] S. Han, S. Lee, D. Kim, J. Seong, A. Sharma, J. Lim, S. Jeong, **G. Zhao**, Y. Chen, S. B. Baek, et al., “Construction of Chimeric Metal–Organic Frameworks with Symmetry-Mismatched Building Blocks”, *Chemistry of Materials*, vol. 35, no. 15, pp. 5903–5913, 2023.
- [18] **G. Zhao**, Q. Zheng, X. Zhang, and W. Zhang, “Adsorption Equilibrium and the Effect of Honeycomb Heat Exchanging Device on Charge/Discharge Characteristic of Methane on MIL-101 (Cr) and Activated Carbon”, *Chinese Journal of Chemical Engineering*, vol. 28, no. 7, pp. 1964–1972, 2020.
- [19] X. Zhang, Q. Zheng, **G. Zhao**, and W. Zhang, “Adsorption Equilibrium and Charge/Discharge Characteristics of Hydrogen on MOFs”, *Cryogenics*, vol. 112, p. 103 121, 2020.
- [20] X. Zhang, Q. Zheng, W. Zhang, and **G. Zhao**, “Study of Adsorption of Ship’s Bilge Oily Water on Expanded Graphite/Activated Carbon Composite Adsorbents”, *Ship Science and Technology*, vol. 42, no. 6, pp. 105–109, 2020.
- [21] **G. Zhao**, Q. Zheng, W. Zhang, and X. Zhang, “Adsorption Equilibrium and Charge/Discharge Characteristics of Methane on MIL-101”, *Journal of Fuel Chemistry and Technology*, vol. 47, pp. 1529–1936, 2019.

DEVELOPMENT

- **MOFRefiner** Generative-AI-Based Models for Unreasonable MOFs Refinement developing
• *Github*
- **MOFAgent** An LLM Agent for End-to-End Computational Materials Discovery model & tool
• *Github*
- **MOF-Sleuth** Tool-Grounded Reward Alignment for Explainable Fine-Grained MOF CIF Auditing model & tool
• *Github*
- **MOF2Text** A Tool Used for Converting MOFs to Text Descriptors. structure & package
• *Github*
- **GTsR** A GNN Based Tool for Solvent Removal from MOF with Stability Check model & package
PyPi • *Github* • *Website* • *Zenodo*
- **MOFClassifier2** Chemically Meaningful Textualization Enables Explainable Validation of Metal-Organic Frameworks by Large Language Models model & package
hugging face • *Github* • *Zenodo*
- **CoRE MOF** Computation-Ready Experimental Metal-Organic Framework database & package
PyPi • *Github* • *Website* • *Zenodo*
- **PACMAN-charge** Partial Atomic Charges for Porous Materials based on GCN model & package
PyPi • *Github* • *Website*
- **MOFClassifier** A Machine Learning Approach for Validating Computation-Ready MOFs model & package
PyPi • *Github*
- **ToBaCCo-MOF** Topologically Based Crystal Constructor package (a refactored version)
PyPi
- **SESAMI** Characterization Tools for Porous Materials Using N_2 /Ar Adsorption Isotherms package (a refactored version)
PyPi

TEACHING

- **Teaching Assistant** at Pusan National University
Introduction to Computational Nano Science and Engineering (CG3500768-062) Spring 2025
- **Teaching Assistant** at Pusan National University
Current Events in Chemical and Biomolecular Engineering (CG1501061-062) Fall 2024
- **Teaching Assistant** at Pusan National University
AI for Chemical and Materials (CG2001538-062) Spring 2024
- **Teaching Assistant** at Pusan National University
Introduction to Data Science for Chemical Engineers (CG3500748-062) Fall 2022

SKILLS

- **Machine Learning Tools:** PyTorch, Scikit-learn, XGBoost, Pymatgen, ASE, RDKit, MLPs
- **Density Functional Theory:** VASP, CP2K, Gaussian
- **Molecular Simulation:** LAMMPS, RASPA, Materials Studio
- **Visualization:** VESTA, Ovito, Avogadro, Matplotlib, Seaborn
- **Programming Languages:** Python, MATLAB, Shell, Markdown, Julia
- **Languages:** Chinese (Native), English (Professional Knowledge), Korean (Basic Knowledge)

RESEARCH INTERESTS

- **AI4Science:** Accelerating materials discovery using artificial intelligence techniques.
- **Big Data:** Using natural language processing and large language model to construct MOF database.
- **High-Throughput Screening:** Multi-scale screening for gas storage and separation.

SCHOLARSHIPS AND AWARDS

- CBE FOCUS (JACS work) in School of Chemical Engineering at Pusan National University 2025
- CBE FOCUS (Matter work) in School of Chemical Engineering at Pusan National University 2025
- PNU Research (Matter work) at Pusan National University 2025
- Featured in the JCTC editorial on machine learning and statistical mechanics for contributing to next-generation chemical theory integration (JCTC work, Cover) 2025
- Excellent Poster Award (KSIEC-Busan) 2024
- Excellent Poster Award (KICChE-Busan) 2024
- Published as part of the Industrial & Engineering Chemistry Research (IECR work) virtual special issue “2023 Class of Influential Researchers” 2023
- Excellent Poster Award (KICChE-Busan) 2023
- Excellent Poster Award (KSIEC-Busan) 2023
- Excellent Poster Award (KSIEC-Busan) 2021
- Foreign Student Scholarship 2020–2021
- Excellent Professional Master’s Degree Thesis Award in Fujian Province 2020
- Excellent Master’s Thesis Award of Jimei University 2020
- Excellent Master’s Degree Graduate 2020
- M.S. Scholarship 2018–2019

- B.S. Scholarship 2017–2018
- Excellent Undergraduate Thesis Award 2018

DOMESTIC / INTERNATIONAL COLLABORATORS

- North America:
 - *Snurr Group @ Northwestern*
 - *HJK Group @ MIT*
 - *Rosen Research Group @ Princeton*
 - *Siepmann Group @ Minnesota*
 - *Sholl Group @ Georgia Tech*
 - *Moosavi Group @ Toronto*
 - UCB, UCLA, CMU, Brown, ...
- Europe:
 - *FX Coudert Group @ PSL University*
 - *Haranczyk @ IMDEA*
 - *Thijs Group @ TU Delft*
- Asia:
 - *Song Li @ HUST*
 - *Koh @ KAIST*
 - *Lah Group @ UNIST*
 - *Hong Group @ KU*
 - *Lin Group @ NTU & OSU*
 - Northwestern Polytechnical University, Beihang, Beijing Institute of Technology, Chinese Academy of Sciences, ...